

# Running & Squeeze Cementing Procedure - ACT-CR-5500 Cement Retainer

Controlled Procedure RSP-CR-5500 Rev B | Product: ACT-CR-5500

|                |                   |
|----------------|-------------------|
| Prepared By    | A. Gutierrez      |
| Checked By     | S. Nguyen         |
| Approved By    | J. Martinez, P.E. |
| Effective Date | 2024-09-02        |

## 1. Scope

Wireline setting of the ACT-CR-5500 cement retainer and subsequent tubing-conveyed squeeze operation with the ACT-SRT-55 stinger.

## 2. Reference Documents

- BOM-CR-5500 Rev B
- DS-CR-5500
- MAN-BP-SER (setting steps)
- API RP 65-2

## 3. Equipment & Materials Required

| Equipment / Material               | Requirement                       |
|------------------------------------|-----------------------------------|
| ACT-ST-10 setting tool + AK-55 kit | Redressed, F-82 current           |
| ACT-SRT-55 stinger assembly        | New chevron seals fitted          |
| Cement unit w/ recorder            | Calibrated                        |
| Slickline gauge cutter run         | Confirm tubing drift 2.00 in. min |

## 4. Procedure

Steps marked 'QC HOLD' are mandatory hold points: work shall not proceed past a hold point without sign-off by the responsible party in the right-hand column. Record actual values on the job traveler.

### 4.1 Wireline Setting

| Step | Action                                                                               | Responsible / Sign-off |
|------|--------------------------------------------------------------------------------------|------------------------|
| S1   | Set retainer per MAN-BP-SER Section 3 at design depth; shear-out 38,000 lbf +/- 10%. | WL Eng                 |
| S2   | Pressure test casing above retainer to 1,000 psi below casing rating, 15 min.        | Co. Man                |

### 4.2 Squeeze Operation

| Step | Action                                                                            | Responsible / Sign-off |
|------|-----------------------------------------------------------------------------------|------------------------|
| Q1   | RIH with SRT-55 stinger on tubing; tag retainer, pick up 2 ft, record weight.     | Operator               |
| Q2   | Sting in: 5,000-8,000 lbf down. Valve opens - establish injection at < 2 bbl/min. | Co. Man                |

| Step | Action                                                                               | Responsible / Sign-off |
|------|--------------------------------------------------------------------------------------|------------------------|
| Q3   | Perform injection test per squeeze design; record rates/pressures.                   | Cementer               |
| Q4   | Mix and pump cement per program; displace to stinger depth + calculated shoe factor. | Cementer               |
| Q5   | Squeeze to design pressure (max 5,000 psi pump-through rating).                      | Cementer               |
| Q6   | Sting out (straight pull); valve closes trapping squeeze pressure below.             | Operator               |
| Q7   | Reverse circulate excess cement above retainer, min 1.5 tubing volumes.              | Cementer               |
| Q8   | POOH; WOC per program; drill out retainer if required (Class 40 cast iron).          | Operator               |

## 5. Notes & Cautions

- Never rotate tubing while stung into the retainer.
- If squeeze pressure exceeds 5,000 psi with the stinger latched, bleed back immediately - the pump-through valve is the limiting component.
- Retainer holds 10,000 psi from below after sting-out.

## 6. Sign-off Record

| Role         | Name              | Signature               | Date       |
|--------------|-------------------|-------------------------|------------|
| Prepared     | A. Gutierrez      | /signed electronically/ | 2024-09-02 |
| Checked (QA) | S. Nguyen         | /signed electronically/ | 2024-09-02 |
| Approved     | J. Martinez, P.E. | /signed electronically/ | 2024-09-02 |